

Revised Load Interconnection System Impact Study Report

For Queue Project Q22-2074

Greenly 230 kV
March 18, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2074. The Interconnection Customer proposes a 219.4 MW large load (data center) facility interconnecting at the Greenly 230 kV substation in the Ohio area. The request was received on April 30, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 501 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,574,000.

3. Network Impacts

3.1 Load Deliverability Analysis

- Facility: John C. Eisinger – E Cortez 100.0 kV
Contingency: Greenly 500 kV/1 kV Transformer
Loading: 108.4% (Rating: 126 MVA)
DFAX: 10.5%
Pre-project loading: 90.1%
- Facility: John C. Eisinger 100/1 kV TX
Contingency: Greenly 230/1kV Transformer Bank
Loading: 100.9% (Rating: 316 MVA)
DFAX: 7.2%
Pre-project loading: 95.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
E Cortez – John C. Eisinger 100.0 kV Line	Reconductor line	\$3,245,000	73% (\$2,356,000)
John C. Eisinger 100/1 kV TX	Replace transformer with higher-rated unit	\$3,218,000	100% (\$3,218,000)
	Total allocated to project		\$5,574,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.